Kanatchikov's quantization: problems and perspectives

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Abstract

In this talk I explain, in a self-contained way, Kanatchikov's approach to field quantization. I introduce the graded Poisson brackets and apply the Dirac algorithm of quantization to obtain the operators associated with each classical observable. I discuss several problems of Kanatchikov's quantization map, including the emergence of non-commutative and non-associative algebras. Finally, I explore a systematic approach based on geometric quantization, leading to higher geometric settings in which quantization becomes possible.

1 Multisymplectic, brackets and canonical relations

The dynamical equations of a field theory, with fields $y^i(x^\mu)$ and lagrangian density $\mathcal{L}$, are given by the Euler-Lagrange equations:

$$\frac{\partial \mathcal{L}}{\partial y^i} - \frac{\partial}{\partial x^\mu}\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu y^i)}\right) = 0 \tag{1}$$

In the hamiltonian multisymplectic formalism, we define the multimomenta as:

$$p_i^\mu \equiv \frac{\partial \mathcal{L}}{\partial(\partial_\mu y^i)} \tag{2}$$

so that there are several momenta for each field, corresponding to each of the directions in spacetime.

The space (y^i, p_i^μ, x^μ) is the multisymplectic phase space, which has a canonical structure given by a closed, non degenerate $(n+1)-$form:

$$\Omega = dy^i \wedge dp_i^\mu \wedge d^{n-1}x_\mu \tag{3}$$

(For this discussion, I will work with this differential form, but if we want to include the dynamics we can add the $dp \wedge d^n x$ term and the constraint $p = -\mathcal{H}$)

This multisymplectic form allows us to establish a mapping between certain observables (differential forms that can be integrated over hypersurfaces) and multivectors via:

$$\iota_{X_F}\Omega = dF \tag{4}$$

if F is an $f-$form, then X_F is an $(n-f)-$multivector.

Multivectors have a natural algebraic structure when endowed with the Schouten-Nijenhuis bracket (which is a generalization of the Lie bracket of vectors). By doing this association, observables can inherit this algebra structure by defining the Poisson bracket as:

$$\iota_{-[X_F, X_G]}\Omega = d\{F, G\} \tag{5}$$

The bracket can be explicitly computed as[1]:

$$\{F, G\} = (-1)^{n-f-1}\iota_{X_G}\iota_{X_F}d\Omega = (-1)^{n-f-1}\iota_{X_G}dF = (-1)^{f-g}\mathcal{L}_{X_G}F \tag{6}$$

We can compute this bracket for several fundamental observables to describe the canonical relations:

$$\{y^i, y^j\} = 0, \qquad \{p_i^\mu d^{n-1}x_\mu, p_j^\nu d^{n-1}x_\mu\} = 0, \qquad \{y^i, p_j^\mu d^{n-1}x_\mu\} = \delta_j^i \tag{7}$$

$$\{y^i d^{n-1}x_\mu, y^j d^{n-1}x_\nu\} = 0, \qquad \{p_i^\mu, p_j^\nu\} = 0, \qquad \{p_i^\mu, y^j d^{n-1}x_\nu\} = \delta_i^j \delta_\nu^\mu \tag{8}$$

2 Kanatchikov quantization

To quantize, we are going to apply the Dirac quantization algorithm[2]:

$$[q(y^i), q(p^\mu_j d^{n-1}x_\mu)] = i\hbar\delta^i_j, \qquad [q(p^\mu_i), q(y^j d^{n-1}x_\nu)] = i\hbar\delta^j_i\delta^\mu_\nu \tag{9}$$

and we will work in the field representation, so that the operator $\hat{y}^i$ acts simply as multiplication by y^i.

In analogy to quantum mechanics, we know that the first commutator implies the following:

$$q(p^\mu_i d^{n-1}x_\mu) = -i\hbar\frac{\partial}{\partial y^i} \tag{10}$$

Now we will find the expression for $q(p^\mu_i)$ and $q(d^{n-1}x_\mu)$ separately. For that, we are going to assume that $q(p^\mu_i d^{n-1}x_\mu) = q(p^\mu_i)q(d^{n-1}x_\mu)$, and the same for $q(y^i d^{n-1}x_\mu) = q(y^i)q(d^{n-1}x_\mu)$.

First problem: Why should we split the operator $q(p^\mu_j d^{n-1}x_\mu)$ into $q(p^\mu_j)q(d^{n-1}x_\mu)$? These are three different observables and therefore should correspond to different operators. Quantization is not expected to preserve the product, see for example the approach of geometric quantization.

Let's admit that the splitting can be done and compute the values of $q(p^\mu_i)$ and $q(d^{n-1}x_\mu)$ using the other commutator. For that, we make the following ansatz for the form of the operator $q(p^\mu_j)$:

$$q(p^\mu_j) = i\hbar\alpha^\mu\partial_i \tag{11}$$

Second problem: This ansatz may not be the one needed to quantize the full algebra of observables, even though it works for these particular ones.

Substituting the ansatz in the commutator:

$$\begin{aligned}
[i\hbar\alpha^\mu\partial_i, y^j q(d^{n-1}x_\nu)]\psi &= i\hbar\alpha^\mu\partial_i\left(y^j q(d^{n-1}x_\nu)\psi\right) - i\hbar y^j q(d^{n-1}x_\nu)\alpha^\mu\partial_i\psi \\
&= i\hbar\alpha^\mu\delta^j_i q(d^{n-1}x_\nu)\psi + i\hbar\alpha^\mu y^j q(d^{n-1}x_\nu)\partial_i\psi - i\hbar y^j q(d^{n-1}x_\nu)\alpha^\mu\partial_i\psi \\
&= i\hbar\alpha^\mu\delta^j_i q(d^{n-1}x_\nu)\psi + i\hbar y^j\left(q(d^{n-1}x_\nu)\alpha^\mu - \alpha^\mu q(d^{n-1}x_\nu)\right)\partial_i\psi
\end{aligned} \tag{12}$$

And given that this has to be equal to multiplication by $i\hbar\delta^i_j\delta^\nu_\mu$, we make the following two equalities:

$$\alpha^\mu q(d^{n-1}x_\nu) = \delta^\mu_\nu, \qquad q(d^{n-1}x_\nu)\alpha^\mu - \alpha^\mu q(d^{n-1}x_\nu) = 0 \tag{13}$$

Therefore we are searching for values such that they commute and the multiplication gives the delta. Here Kanatchikov proposes the values for α^μ and $q(d^{n-1}x_\nu)$ in a Clifford algebra, so that they fulfil these expresions:

$$\alpha^\mu = \kappa\gamma^\mu, \qquad q(d^{n-1}x_\mu) = \frac{1}{\kappa}\gamma_\mu \tag{14}$$

So that, finally, the three operators are:

$$q(y^i) = y^i, \qquad q(p^\mu_i) = i\hbar\kappa\gamma^\mu\frac{\partial}{\partial y^i}, \qquad q(d^{n-1}x_\mu) = \frac{1}{\kappa}\gamma_\mu \tag{15}$$

2.1 Why Kanatchikov quantization is not correct and how it can be fixed

Third problem: This quantization does not hold.

This is easy to see. Clifford algebra is not commutative. Therefore, if we check the second condition for the algebra:

$$\begin{aligned}
q(d^{n-1}x_\nu)\alpha^\mu - \alpha^\mu q(d^{n-1}x_\nu) &= (\kappa\gamma^\mu)\left(\frac{1}{\kappa}\gamma_\nu\right) - \left(\frac{1}{\kappa}\gamma_\nu\right)(\kappa\gamma^\mu) \\
&= \gamma^\mu\gamma_\nu - \gamma_\nu\gamma^\mu \\
&= -2\gamma_\nu\gamma^\mu + 2\delta^\mu_\nu
\end{aligned} \tag{16}$$

which is nonzero, so the Clifford algebra does not satisfy the algebra conditions.

However, if we take the symmetrization of this Clifford algebra, both equations are fulfilled. Let us denote by $\odot$ the operator that ONLY symmetrizes the Clifford algebra, and not the rest of the differential operator. Then:

$$\begin{aligned}\alpha^{\mu} \odot q(d^{n-1}x_{\nu}) &= \gamma^{\mu} \odot \gamma_{\nu} \\ &= \frac{1}{2}(\gamma^{\mu}\gamma_{\nu} + \gamma_{\nu}\gamma^{\mu}) = \delta^{\mu}_{\nu}\end{aligned} \tag{17}$$

and, by definition of the symmetrization, the second algebra condition also holds.

This is the Jordan algebra associated with the Clifford algebra. Note that this Jordan algebra is not associative:

$$a \odot (b \odot c) = \frac{1}{2} a \odot (bc + cb) = \frac{1}{4}(abc + acb + bca + cba) \tag{18}$$

$$(a \odot b) \odot c = \frac{1}{2}(ab + ba) \odot c = \frac{1}{4}(abc + bac + cab + cba) \tag{19}$$

therefore $a \odot (b \odot c) \neq (a \odot b) \odot c$.

Theorem: For $n \geq 2$, it does not exist any associative algebra satisfying the following two conditions:

$$\widehat{p^{\mu}} \circ \widehat{\omega_{\nu}} = \delta^{\mu}_{\nu}, \qquad \widehat{p^{\mu}} \circ \widehat{\omega_{\nu}} - \widehat{\omega_{\nu}} \circ \widehat{p^{\mu}} = 0 \tag{20}$$

Proof. We can prove this by contradiction. By hypothesis, if we take the index $i = j$ in the first equation, we obtain $\widehat{p^i} \circ \widehat{\omega_j} = 1$. Now, we can multiply this equality on the left by some $\widehat{p^k}$, with $k \neq i$ and $k \neq j$, obtaining $\widehat{p^k} \circ \widehat{p^i} \circ \widehat{\omega_j} = \widehat{p^k}$. But since $\widehat{p^i}$ and $\widehat{\omega_j}$ commute, by the second equation, the equality can be written as $\widehat{p^k} \circ \widehat{\omega_j} \circ \widehat{p^i} = \widehat{p^k}$. However, since $k \neq j$, by the first equation the product on the left-hand side is zero, which implies that $\widehat{p^k} = 0$. But this argument can be repeated for all indices k, so that $\widehat{p^{\mu}} = 0$, which contradicts the initial assumption, namely, that $\widehat{p^{\mu}} \circ \widehat{\omega_{\nu}} = \delta^{\mu}_{\nu}$. □

First solution: redefine the commutators with the Clifford algebra symmetrization product. In this way, the commutators will be defined in this way:

$$[A, B] = A \odot B - B \odot A \tag{21}$$

In this way, the quantized operators that Kanatchikov proposes fulfil the canonical commutator relations.

For me, this is kinda arbitrary, so I prefer the second solution.

Second solution: redefine the Clifford algebra elements with the symmetrization product inside, i.e., the algebra elements act by symmetrizing whatever they acts on:

$$q(y^i) = y^i, \qquad q(p^{\mu}_i) = i\hbar\kappa\gamma^{\mu} \odot \frac{\partial}{\partial y^i}, \qquad q(d^{n-1}x_{\mu}) = \frac{1}{\kappa}\gamma_{\mu}\odot \tag{22}$$

For Kanatchikov, however, this cannot be, because $\gamma^{\mu}\gamma^{\nu}$ seems to be important in his theory of quantum gravity.

This is equivalent to, rather than working with the Clifford algebra elements, working with its symmetrization, leading to a Jordan algebra:

$$\gamma_{\mu} \odot \gamma^{\nu} = \frac{\gamma_{\mu}\gamma^{\nu} + \gamma^{\nu}\gamma_{\mu}}{2} \tag{23}$$

Fourth problem: The metric, and therefore the Clifford algebra, is not a canonical structure of the multisymplectic phase space, and should not be used for quantization.

2.2 Examples

2.2.1 Scalar field

The hamiltonian of the free scalar field with mass m is given by:

$$\hat{\mathcal{H}} = -\frac{1}{2}\hbar^2\kappa^2\frac{\partial^2}{\partial\phi^2} + \frac{1}{2}\frac{m^2c^2}{\hbar^2}\phi^2 \tag{24}$$

The eigenvalues and eigenstates of this operator are given by:

$$\hat{\mathcal{H}}_n\psi_n = \eta_n\psi_n, \qquad \psi_n = \frac{1}{\sqrt{2^n n!}}\sqrt[4]{\frac{m}{\pi\hbar\kappa}}\exp\left(-\frac{1}{2}\frac{m}{\hbar\kappa}\phi^2\right)H_n\left(\sqrt{\frac{m}{\hbar\kappa}}\phi\right), \qquad \eta_n = \kappa mc(n + 1/2) \tag{25}$$

2.2.2 Spacetime evolution

The Schrödinger equation is the following:

$$i\hbar\kappa\gamma^{\mu} \odot \partial_{\mu}\Psi = \hat{\mathcal{H}} \odot \Psi \tag{26}$$

If we assume that the DW Hamiltonian does not depend on the spacetime coordinates explicitly, we can do variable separation and write:

$$i\hbar\kappa\gamma^{\mu} \odot \partial_{\mu}\Psi = \eta \odot \Psi \tag{27}$$

where η are the common eigenvalues of the operators of spacetime evolution and the hamiltonian. Note that η is possibly Clifford algebra evaluated. We now try to solve this equation for a given η. We take $\Psi = \psi_{\star} + \psi_{\mu}\gamma^{\mu}$ and $\eta = \eta_{\star} + \eta_{\mu}\gamma^{\mu}$:

$$i\hbar\kappa\partial_{\mu}\left(\gamma^{\mu} \odot (\psi_{\star} + \psi_{\nu}\gamma^{\nu})\right) = (\eta_{\star} + \eta_{\mu}\gamma^{\mu}) \odot (\psi_{\star} + \psi_{\nu}\gamma^{\nu}) \tag{28}$$

The $\odot$ product is distributive, so we can write this equation as:

$$i\hbar\kappa\partial_{\mu}(\psi_{\star}\gamma^{\mu} + \psi_{\nu}\gamma^{\mu} \odot \gamma^{\nu}) = \eta_{\star}\psi_{\star} + \eta_{\star}\psi_{\mu}\gamma^{\mu} + \psi_{\star}\eta_{\mu}\gamma^{\mu} + \eta_{\mu}\psi_{\nu}\gamma^{\mu} \odot \gamma^{\nu} \tag{29}$$

And using $\gamma^{\mu} \odot \gamma^{\nu} = \eta^{\mu\nu}$:

$$i\hbar\kappa\partial_{\mu}\psi_{\star}\gamma^{\mu} + i\hbar\kappa\partial_{\mu}\psi^{\mu} = \eta_{\star}\psi_{\star} + \eta_{\mu}\psi^{\mu} + (\psi_{\star}\eta_{\mu} + \eta_{\star}\psi_{\mu})\,\gamma^{\mu} \tag{30}$$

so that, equating both sides termwise, we obtain the following system of differential equations:

$$\begin{cases} i\hbar\kappa\partial_{\mu}\psi^{\mu} = \eta_{\star}\psi_{\star} + \eta_{\mu}\psi^{\mu} \\ i\hbar\kappa\partial_{\mu}\psi_{\star} = \eta_{\star}\psi_{\mu} + \psi_{\star}\eta_{\mu} \end{cases} \tag{31}$$

This system of equations can be written in terms of Duffin-Kemmer-Petiau matrices:

$$(i\hbar\kappa\beta^{\mu}\partial_{\mu} - E)\Psi = 0 \tag{32}$$

where:

$$\beta^0 = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \beta^1 = \begin{pmatrix} 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \beta^2 = \begin{pmatrix} 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \tag{33}$$

$$\beta^3 = \begin{pmatrix} 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}, \qquad E = \begin{pmatrix} \eta_{\star} & \eta_0 & \eta_1 & \eta_2 & \eta_3 \\ \eta_0 & \eta_{\star} & 0 & 0 & 0 \\ \eta_1 & 0 & -\eta_{\star} & 0 & 0 \\ \eta_2 & 0 & 0 & -\eta_{\star} & 0 \\ \eta_3 & 0 & 0 & 0 & -\eta_{\star} \end{pmatrix} \tag{34}$$

It is interesting to note that this Duffin-Kemmer-Petiau matrices and the corresponding equation were already created in 1930s decade to model scalars and vector quantum fields simultaneously. However, the definition of this algebra of matrices uses the metric:

$$\beta^{\mu}\beta^{\nu}\beta^{\rho} + \beta^{\rho}\beta^{\nu}\beta^{\mu} = \beta^{\mu}\eta^{\nu\rho} + \beta^{\rho}\eta^{\nu\mu} \tag{35}$$

A possible relation here with the canonical form (so that we don't use the metric in the quantum field theory formulation) is that, only in the case of a euclidean signature, and using $Z^i \equiv (\phi, p^0, p^1, p^2, p^3)$ as local coordinates, the Hamilton-De Donder-Weyl equations of motion can be written in terms of this matrices[3]:

$$\beta^{\mu}_{ij}\partial_{\mu}Z^j = \frac{\partial H}{\partial Z^i} \tag{36}$$

which is analogous to what can be done with the symplectic form using $z = (q, p)$ and $\omega\partial_t z = \partial_z H$.

The general non-trivial solution of the system of differential equations is:

$$\Psi_k = \begin{cases} A\exp\left(\frac{\eta_\mu x^\mu}{i\hbar\kappa}\right) + n_\mu\gamma^\mu \exp\left(\frac{\eta_\nu x^\nu}{i\hbar\kappa}\right) & \text{if } \eta_\star = 0 \\ A\left(\exp\left(ik_\mu x^\mu\right) - \frac{\hbar\kappa k_\mu + \eta_\mu}{\eta_\star}\gamma^\mu \exp\left(ik_\mu x^\mu\right)\right) & \text{if } \eta_\star \neq 0 \end{cases} \tag{37}$$

where n_μ is any divergenceless vector and k_μ is any vector fulfilling the following condition:

$$\eta_\star^2 = \left(\hbar\kappa k_\mu + \eta_\mu\right)^2 \tag{38}$$

2.2.3 Full expression of the vacuum scalar field wave function

For the scalar field, we found previously the eigenvalues of the hamiltonian, which is the right-hand side of the Schrödinger equation. Therefore $\eta_n = \kappa mc(n + 1/2)$. This solution has no Clifford part, so the only contribution is the one of $\eta_\star$:

$$\psi_0 = \sqrt[4]{\frac{m}{\pi\hbar\kappa}} \exp\left(-\frac{1}{2}\frac{m}{\hbar\kappa}\phi^2\right), \qquad \eta = \frac{1}{2}\kappa mc \tag{39}$$

The full expression of the wave function will be the product of the spacetime evolution wave function and the field part. Recall that the spacetime evolution is given by:

$$\Psi_k(x^\mu) = A\left(\exp\left(ik_\mu x^\mu\right) - \frac{\hbar\kappa k_\mu + \eta_\mu}{\eta_\star}\gamma^\mu \exp\left(ik_\mu x^\mu\right)\right), \qquad \text{where } \eta_\star^2 = \left(\hbar\kappa k_\mu + \eta_\mu\right)^2 = 0 \tag{40}$$

In this case, $\eta_\mu = 0$, so, $k_\mu k^\mu = \left(\frac{\eta_\star}{\hbar\kappa}\right)^2$, with $\eta_\star = \frac{1}{2}\kappa mc$, and the wave function simplifies to:

$$\Psi_u(x^\mu) = A\left(\exp(ik_\mu x^\mu) - \frac{2\hbar}{mc}k_\mu\gamma^\mu \exp(ik_\mu x^\mu)\right), \qquad \text{where } k_\mu k^\mu = \left(\frac{mc}{2\hbar}\right)^2 \tag{41}$$

We can write this in a more simplified version if we use $u_\mu \equiv \frac{2\hbar}{mc}k_\mu$ is some unit vector:

$$\Psi_u(x^\mu) = A\left(\exp\left(i\frac{mc}{2\hbar}u_\mu x^\mu\right) - u_\mu\gamma^\mu \exp\left(i\frac{mc}{2\hbar}u_\mu x^\mu\right)\right) \tag{42}$$

The full expression of the wave function for the vacuum is:

$$\Psi_{u,0}(\phi, x^\mu) = B\left(\exp\left(i\frac{mc}{2\hbar}u_\mu x^\mu\right) - u_\mu\gamma^\mu \exp\left(i\frac{mc}{2\hbar}u_\mu x^\mu\right)\right)\exp\left(-\frac{1}{2}\frac{m}{\hbar\kappa}\phi^2\right) \tag{43}$$

where we have reabsorbed the term $\sqrt[4]{\frac{m}{\pi\hbar\kappa}}$ into the multiplicative constant.

2.3 Problems of non commutative and non associative algebras for quantum mechanics

Fifth problem: Diagonalization of matrices evaluated over Clifford algebras is non-unique.

To see this, the first thing to ask is whether, over a non-commutative ring, the left inverse of a matrix is the same as its right inverse. The answer to this question is that if a matrix A admits a left inverse such that $A_L^{-1}A = \mathbb{I}$ and a right inverse such that $AA_R^{-1} = \mathbb{I}$, then both inverses are equal, as can be verified below:

$$A_L^{-1} = A_L^{-1}(AA_R^{-1}) = (A_L^{-1}A)A_R^{-1} = A_R^{-1} \tag{44}$$

Sixth problem: This is no longer true in the Jordan algebra of the symmetrized Clifford algebra. In this setup, diagonalization makes even less sense.

Now, if we want to write a matrix M in the form PDP^{-1}, the matrix D will correspond to the right eigenvalues of the matrix, that is, $Mv_i = v_i\lambda_i$, while the change-of-basis matrix P will consist of the matrix whose columns are the eigenvectors associated with the eigenvalues appearing in D. We can verify this as follows: suppose there exist P and D such that $M = PDP^{-1}$, where D is a diagonal matrix with elements λ_i, meaning that $De_i = \lambda_i e_i$. We define $v_i = Pe_i$. Then, by multiplying M by v_i, we obtain:

$$\begin{aligned} Mv_i &= PDP^{-1}v_i \\ &= PDP^{-1}(Pe_i) \\ &= PDe_i \\ &= P\lambda_i e_i \\ &= Pe_i\lambda_i \\ &= v_i\lambda_i \end{aligned} \tag{45}$$

That is, the elements of the diagonal correspond to the right eigenvalues $Mv_i = v_i\lambda_i$, as we wanted to prove.

A difference compared to what happens in linear algebra is that, if a vector v_i is an eigenvector with eigenvalue λ_i of a matrix M, then αv_i is not necessarily another eigenvector of M. That is, $\operatorname{span}\{v_i\}$ is not necessarily an invariant subspace. Instead, we have the following two theorems:

Theorem: If v_i is an eigenvector of M over an R-module with eigenvalue $\lambda_i \in Z[R]$, then $v_i\alpha$ is also an eigenvector of M for all $\alpha \in R$, and $\operatorname{span}\{v_i\}$ is an invariant subspace.[1]

Theorem: If v_i is an eigenvector of M over an R-module with eigenvalue λ_i, then $\alpha v_i = v_i\alpha$ is also an eigenvector of M for all $\alpha \in Z[R]$, and $\{\alpha v_i : \alpha \in Z[R]\}$ is an invariant subspace.

Proof: By the hypothesis of the theorem, $Mv_i = v_i\lambda_i$. If we multiply on the right by α, we obtain $Mv_i\alpha = v_i\lambda_i\alpha$. If λ_i commutes with all elements of R, then $M(v_i\alpha) = (v_i\alpha)\lambda_i$, which proves the first theorem. If, on the other hand, it is α that commutes with all elements of R, then $M(v_i\alpha) = (v_i\alpha)\lambda_i = (\alpha v_i)\lambda_i$, which proves the second theorem.

Example: Let us compute the right eigenvalues of the following matrix:

$$M = \begin{pmatrix} 0 & e_0 \\ e_0 & 0 \end{pmatrix} \tag{46}$$

We seek eigenvalues and eigenvectors $Mv_i = v_i\lambda_i$ of the form:

$$\begin{pmatrix} 0 & e_0 \\ e_0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x\lambda \\ y\lambda \end{pmatrix} \tag{47}$$

That is, we want to solve the following system of equations:

$$\begin{cases} e_0 y = x\lambda \\ e_0 x = y\lambda \end{cases} \tag{48}$$

Multiplying the first equation by e_0 on the left we obtain $y = e_0 x\lambda$, and substituting into the second equation:

$$e_0 x = e_0 x\lambda^2 \Longrightarrow \lambda^2 = 1 \tag{49}$$

The equation $\lambda^2 = 1$ has uncountable many solutions in Clifford algebra, so the matrix has infinitely many right eigenvalues $Mv_i = v_i\lambda_i$, including $\lambda = \pm 1$, $\lambda = \pm e_0$, $\lambda = \pm e_1$, and others of the form $be_0 + ce_1 + de_0e_1$ where $b^2 + c^2 - d^2 = 1$, such as $\lambda = e_0 + e_1 + e_0e_1$ or $\lambda = 5e_0 + 5e_1 + 7e_0e_1$.

As we have mentioned, the right eigenvalues coincide with the diagonal form of the matrix, so it can be diagonalized as $P^{-1}DP$ in different ways:

$$\begin{gathered} M = \begin{pmatrix} 1/2 & e_0/2 \\ 1/2 & -e_0/2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ e_0 & -e_0 \end{pmatrix}, \qquad M = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & e_0e_1/2 \end{pmatrix} \begin{pmatrix} e_0 & 0 \\ 0 & e_1 \end{pmatrix} \begin{pmatrix} 1 - e_0e_1 & 1 + e_0e_1 \\ 1 + e_0e_1 & -1 - e_0e_1 \end{pmatrix}, \\ M = \begin{pmatrix} 1 & 1 \\ 1 + e_1 + e_0e_1 & 5 + 7e_1 + 5e_0e_1 \end{pmatrix} \begin{pmatrix} e_0 + e_1 + e_0e_1 & 0 \\ 0 & 5e_0 + 5e_1 + 7e_0e_1 \end{pmatrix} \begin{pmatrix} 1/2 - e_0/2 + e_1/2 & 1 - \frac{3}{2}e_1 - e_0e_1 \\ 1/2 + e_0/2 - e_1/2 & -1 + \frac{3}{2}e_1 + e_0e_1 \end{pmatrix} \end{gathered} \tag{50}$$

[1] $Z[R]$ denotes the center of the ring, that is, the set of elements that commute with every element of the ring. In the case of Clifford algebra, these are only the real numbers.

among others.

Note: Not every pair of eigenvalues can be used to write the matrix M in its diagonal form, since some matrices formed by the eigenvectors associated with those eigenvalues might not be invertible. Notice that the vectors do not need to be linearly dependent for the matrix to be non-invertible. For example, the following matrix, corresponding to the eigenvectors for $\lambda = 1$ and $\lambda = e_0$, is not invertible:

$$M = \begin{pmatrix} 1 & 1 \\ 1 & e_0 \end{pmatrix} \tag{51}$$

In general, there is not necessarily an analytic algorithm to find the eigenvalues (and, therefore, the different diagonal forms) of a matrix. In this case, we were able to find them analytically due to the simplicity of the proposed matrix, but let us see what happens with a different matrix.

Example: Let us compute the right eigenvalues of the following matrix:

$$M = \begin{pmatrix} 0 & e_0 \\ 2e_1 & 0 \end{pmatrix} \tag{52}$$

We seek eigenvalues and eigenvectors $Mv_i = v_i\lambda_i$ of the form:

$$\begin{pmatrix} 0 & e_0 \\ 2e_1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x\lambda \\ y\lambda \end{pmatrix} \tag{53}$$

That is, we want to solve the following system of equations:

$$\begin{cases} e_0 y = x\lambda \\ 2e_1 x = y\lambda \end{cases} \tag{54}$$

Multiplying the first equation by e_0 on the left gives $y = e_0 x\lambda$, and substituting into the second equation:

$$2e_1 x = e_0 x\lambda^2 \tag{55}$$

In this case, we cannot cancel x, because it is in the middle of the expression, so we cannot multiply by its inverse from either the left or the right to cancel it.

One thing we can do here is to set $x = 1$ to find the eigenvalue in this case. For example:

$$2e_0e_1 = \lambda^2 \tag{56}$$

Assuming an expression $\lambda = a + be_0 + ce_1 + de_0e_1$, we obtain:

$$\begin{cases} a^2 + b^2 + c^2 - d^2 = 0 \\ 2ab = 0 \\ 2ac = 0 \\ 2ad = 2 \end{cases} \tag{57}$$

Thus, we conclude that $b = c = 0$ and $a^2 = d^2$, hence $a = \pm d$. Finally, we obtain the eigenvalues $\lambda = 1 + e_0e_1$, $\lambda = -1 - e_0e_1$, $\lambda = i - ie_0e_1$, and $\lambda = -i + ie_0e_1$.

However, if we had chosen a different value for x, for example $x = e_0$, we would have obtained a different equation:

$$2e_1e_0 = \lambda^2 \tag{58}$$

leading to the following eigenvalues: $\lambda = 1 - e_0e_1$, $\lambda = -1 + e_0e_1$, $\lambda = i + ie_0e_1$, and $\lambda = -i - ie_0e_1$.

Thus, this matrix can be diagonalized in multiple ways:

$$M = \begin{pmatrix} 1 & 1 \\ e_0 + e_1 & -e_0 - e_1 \end{pmatrix} \begin{pmatrix} 1 + e_0e_1 & 0 \\ 0 & -1 - e_0e_1 \end{pmatrix} \begin{pmatrix} 1/2 & \frac{1}{4}(e_0 + e_1) \\ 1/2 & -\frac{1}{4}(e_0 + e_1) \end{pmatrix} \tag{59}$$

$$M = \begin{pmatrix} e_0 & e_0 \\ 1 - e_0e_1 & -1 + e_0e_1 \end{pmatrix} \begin{pmatrix} 1 - e_0e_1 & 0 \\ 0 & -1 + e_0e_1 \end{pmatrix} \begin{pmatrix} e_0/2 & \frac{1}{4}(1 + e_0e_1) \\ e_0/2 & -\frac{1}{4}(1 + e_0e_1) \end{pmatrix} \tag{60}$$

Or we can even mix the eigenvalues obtained by the two different methods:

$$M = \begin{pmatrix} 1 & e_0 \\ e_0 + e_1 & -1 + e_0 e_1 \end{pmatrix} \begin{pmatrix} 1 + e_0 e_1 & 0 \\ 0 & -1 + e_0 e_1 \end{pmatrix} \begin{pmatrix} 1/2 & \frac{1}{4}(e_0 + e_1) \\ e_0/2 & -\frac{1}{4}(1 + e_0 e_1) \end{pmatrix} \tag{61}$$

Again, the matrix cannot be diagonalized with just any pair of eigenvectors. For example, the eigenvector $\begin{pmatrix} 1 \\ e_0 + e_1 \end{pmatrix}$, with eigenvalue $\lambda = 1 + e_0 e_1$, is proportional to the eigenvector $\begin{pmatrix} e_0 \\ 1 - e_0 e_1 \end{pmatrix}$, with eigenvalue $\lambda = 1 - e_0 e_1$, so the matrix they form is not invertible.

2.4 Casimir effect

Let's study the symmetry of the scalar field under diffeomorphisms $x^\mu \to x^\mu - \xi^\mu(x)$. The vector generating this transformation in the multisymplectic phase space is given by the lift of the vector $-\xi^\mu \partial_\mu$ to the multisymplectic phase space[4]:

$$V_\xi = -\xi^\mu \partial_\mu - (p^\nu \partial_\nu \xi^\mu - p^\mu \partial_\nu \xi^\nu)\, \partial_{p^\mu} \tag{62}$$

The observable associated with this vector field via $\iota_{V_\xi} \Omega = dG_\xi$ is given by:

$$G_\xi = \xi^\mu \left(p^\nu d\phi \wedge d^{n-2} x_{\mu\nu} + \mathcal{H} d^{n-1} x_\mu \right) \tag{63}$$

This observable is related with the energy-momentum tensor of a scalar field. This can be understood if we evaluate this operator on a section $\psi = (\phi(x^\mu), p^\nu(x^\mu), x^\mu)$:

$$\begin{aligned}
\psi^* G_\xi &= \xi^\mu \left(p^\nu \partial_\alpha \phi dx^\alpha \wedge d^{n-2} x_{\mu\nu} + \mathcal{H} d^{n-1} x_\mu \right) \\
&= \xi^\mu \left(p^\nu \partial_\alpha \phi \left(\delta^\alpha_\mu d^{n-1} x_\nu - \delta^\alpha_\nu d^{n-1} x_\mu \right) + \mathcal{H} d^{n-1} x_\mu \right) \\
&= \xi^\mu \left(p^\nu \partial_\mu \phi d^{n-1} x_\nu - p^\alpha \partial_\alpha \phi d^{n-1} x_\mu + \mathcal{H} d^{n-1} x_\mu \right) \\
&= \xi^\mu \left(p^\nu \partial_\mu \phi d^{n-1} x_\nu - \mathcal{L} d^{n-1} x_\mu \right) \\
&= \xi^\mu \left(p^\nu \partial_\mu \phi - \delta^\nu_\mu \mathcal{L} \right) d^{n-1} x_\nu \\
&= \xi^\mu T^\nu_\mu d^{n-1} x_\nu
\end{aligned} \tag{64}$$

where we have defined $T^\nu_\mu \equiv p^\nu \partial_\mu \phi - \delta^\nu_\mu \mathcal{L}$ as the energy-momentum tensor. This differential form can be integrated over a constant time hypersurface Σ to obtain the total energy and momentum of the configuration:

$$Q_\xi = \int \xi^\mu T^0_\mu d^3 x, \qquad E = \int T^0_0 d^3 x, \qquad P_i = \int T^0_i d^3 x \tag{65}$$

To obtain the energy-momentum operator in Kanatchikov's quantization, we compute the Poisson bracket with respect to the other observables of the theory:

$$\begin{aligned}
\{F, G_\xi\} &= \{F, \xi^\mu \left(p^\nu d\phi \wedge d^{n-2} x_{\mu\nu} + \mathcal{H} d^{n-1} x_\mu \right)\} \\
&= \iota_{V_\xi} dF
\end{aligned} \tag{66}$$

For the different basic observables of the scalar field we obtain:

$$\{\phi, G_\xi\} = 0, \qquad \{p^\mu, G_\xi\} = -p^\nu \partial_\nu \xi^\mu - p^\mu \partial_\nu \xi^\nu, \qquad \{x^\mu, G_\xi\} = -\xi^\mu \tag{67}$$

From now on, we will assume that the vector field ξ is constant in the basis we are working, so that $\partial_\nu \xi^\mu = 0$. This is the case for measuring the energy and momentum of the field with the vectors ∂_t and ∂_i. In this case, the graded Poisson brackets are:

$$\{\phi, G_\xi\} = 0, \qquad \{p^\mu, G_\xi\} = 0, \qquad \{x^\mu, G_\xi\} = -\xi^\mu \tag{68}$$

When quantizing the energy-momentum operator we promote the Poisson brackets to commutators of operators:

$$[\hat{x^\mu}, \hat{G}_\xi] = -i\hbar \xi^\mu \tag{69}$$

If we work in the spacetime representation, so that $\hat{x}^\mu = x^\mu$ is just the multiplication, then the energy-momentum operator takes the form:

$$\hat{G}_\xi = i\hbar\xi^\mu \frac{\partial}{\partial x^\mu} \tag{70}$$

Now let's apply this to the case previously computed of the vacuum state of the scalar field:

$$\Psi_u = B\left(\exp\left(i\frac{mc}{2\hbar}u_\mu x^\mu\right) - u_\mu\gamma^\mu \exp\left(i\frac{mc}{2\hbar}u_\mu x^\mu\right)\right)\exp\left(-\frac{1}{2}\frac{m}{\hbar\kappa}\phi^2\right) \tag{71}$$

Applying the operator $\hat{G}_\xi$ on this wavefunction:

$$\hat{G}_\xi\Psi_u = -\frac{1}{2}mcu_\mu\xi^\mu\Psi_u \tag{72}$$

Therefore the zero point wavefunction of the scalar field is also eigenstate of the energy-momentum tensor, with real eigenvalue $-\frac{1}{2}mcu_\mu\xi^\mu$. Note that u_μ is a timelike vector, given that $u_\mu u^\mu = 1$. If we want to compute the total energy of the system, we have to choose another timelike $\xi^\mu = (1, 0, 0, 0)$, so the product $u_\mu\xi^\mu$ is going to be positive and, therefore, the energy:

$$\left\langle p^\nu d\phi \wedge d^{n-2}x_{0\nu} + \mathcal{H}d^{n-1}x_0\right\rangle = -\frac{1}{2}mcu_0 \tag{73}$$

I still haven't done an interpretation of the expected value of a differential form to be a constant. Note, however, that this expected value is negative, which may have implications on Casimir effect.

This is not, however, the expected result. The usual Casimir effect has the following expression for the pressure:

$$P = -\frac{\hbar c\pi^2}{240a^4} \tag{74}$$

In fact, we haven't used in any moment of the derivation that we are between two plates in where the field vanishes. Taking this into account is, however, much more complicated that expected in the multisymplectic formalism. This is because in the usual formulation, the field ϕ is forced to vanish at $x = 0$ and $x = a$. However, this imply knowing the full configuration of the field, both its value and its velocity, which violates the uncertainty principle. An option one can consider within this formalism is to make wave function to vanish at $x = 0$ and $x = a$, appearing as a boundary conditions in the solutions of the integral $\Psi(\phi, t, 0, y, z) = \Psi(\phi, t, a, y, z) = 0$. However, this leads to nothing because imposing this conditions in the Schrödinger-Kanatchikov equation implies that the wave function is zero everywhere, because the left-hand-side of the equation is first order in derivatives, so there is no Sturm-Liouville equation, on contrary to the right-hand-side of the equation.

Making a field to not contribute in the computations after some point of the spacetime is like making a particle disappear at some point of time, and this cannot be modeled within the usual framework of quantum mechanics.

3 A different approach: geometric quantization

3.1 Geometric quantization for particles

Quantum systems can be obtained from classical systems by canonical quantization, which consists in finding a $\mathbb{C}-$linear map[2] $q : obs \to \mathrm{End}(\mathbb{H})$ fulfilling the following two Dirac conditions[5]:

(D1) $q(1) = \mathbb{I}$

(D2) $[q(F), q(G)] = i\hbar q(\{F, G\})$

[2]Here observables are C^∞ functions of the phase space.

where $[\cdot,\cdot]$ is the usual commutator of operators.

In fact, this quantization map q is an algebra morphism between $(obs, \{\cdot,\cdot\})$ and $(\mathrm{End}(\mathbb{H}), [\cdot,\cdot])$.

Ansatz: Let $q(F) \equiv F\mathbb{I} - i\hbar\mathcal{L}_{X_F} - \theta(X_F)\mathbb{I}$, such that it acts on functions $\psi \in \mathbb{H}$ in the following way:

$$[q(F)](\psi) = F\psi - i\hbar\mathcal{L}_{X_F}\psi - \theta(X_F)\psi \tag{75}$$

Theorem: The map $q(F)$ is $\mathbb{C}-$linear, and satisfies (D1). Moreover, it satisfies (D2) iff $\omega = -d\theta$.

Proof: (D1) is trivial because the vector field associated with the constant functions via $\iota_X\omega = df$ is zero, so the only contribution for constant functions is $q(\lambda) = \lambda\mathbb{I}$.

Now we will check (D2). Let $\mu(F) \equiv (F - \theta(X_F))\mathbb{I}$ denote the multiplicative part of the quantization map. Then:

$$[q(F), q(G)] = (-i\hbar)^2[\mathcal{L}_{X_F}, \mathcal{L}_{X_G}] - i\hbar[\mathcal{L}_{X_F}, \mu(G)] - i\hbar[\mu(F), \mathcal{L}_{X_G}] + [\mu(F), \mu(G)] \tag{76}$$

The first bracket:

$$[\mathcal{L}_{X_F}, \mathcal{L}_{X_G}] = \mathcal{L}_{[X_F, X_G]} = -\mathcal{L}_{X_{\{F,G\}}} \tag{77}$$

The second bracket:

$$\begin{aligned}[\mathcal{L}_{X_F}, \mu(G)](\psi) &= \mathcal{L}_{X_F}(\mu(G)\psi) - \mu(G)\mathcal{L}_{X_F}\psi \\ &= \mathcal{L}_{X_F}\mu(G)\psi + \mu(G)\mathcal{L}_{X_F}\psi - \mu(G)\mathcal{L}_{X_F}\psi \\ &= \mathcal{L}_{X_F}\mu(G)\psi\end{aligned} \tag{78}$$

and therefore:

$$\begin{aligned}[\mathcal{L}_{X_F}, \mu(G)] &= \mathcal{L}_{X_F}\mu(G) \\ &= \{\mu(G), F\}\end{aligned} \tag{79}$$

Similarly, the third bracket:

$$[\mu(F), \mathcal{L}_{X_G}] = -[\mathcal{L}_{X_G}, \mu(F)] = -\{\mu(F), G\} \tag{80}$$

And multiplication commutes, so the fourth bracket is zero. The full bracket is therefore:

$$\begin{aligned}[q(F), q(G)] &= -(i\hbar)^2\mathcal{L}_{X_{\{F,G\}}} + i\hbar\{F, \mu(G)\} + i\hbar\{\mu(F), G\} \\ &= i\hbar\left(2\{F,G\} - i\hbar\mathcal{L}_{X_{\{F,G\}}} - \{F, \theta(X_G)\} - \{\theta(X_F), G\}\right)\end{aligned} \tag{81}$$

So, if we want the second Dirac condition to be fulfilled, this has to be equal to the quantization of the Poisson bracket:

$$[q(F), q(G)] = i\hbar\left(\{F,G\} - i\hbar\mathcal{L}_{X_{\{F,G\}}} - \theta(X_{\{F,G\}})\right) \tag{82}$$

This is only true if and only if:

$$-\theta(X_{\{F,G\}}) = \{F,G\} - \{F, \theta(X_G)\} - \{\theta(X_F), G\} \tag{83}$$

or, reordering:

$$\{F,G\} = -\{\theta(X_G), F\} + \{\theta(X_F), G\} + \theta([X_F, X_G]) \tag{84}$$

Now we can use the following formula the Poisson brackets:

$$\{F,G\} = \omega(X_F, X_G) \tag{85}$$

and the following formula for the exterior derivative of a 1-form acting in two vectors:

$$\begin{aligned}d\theta(X_F, X_G) &= \mathcal{L}_{X_F}\theta(X_G) - \mathcal{L}_{X_G}\theta(X_F) - \theta([X_F, X_G]) \\ &= \{\theta(X_G), F\} - \{\theta(X_F), G\} - \theta([X_F, X_G])\end{aligned} \tag{86}$$

Therefore, we can rewrite eq. (84) with eqs. (85) and (86) as:

$$\omega(X_F, X_G) = -d\theta(X_F, X_G) \tag{87}$$

Therefore the form θ has to be a symplectic potential. $\square$

Problem: This map is not canonical because it depends on the choice of symplectic potential.

Problem: The operators obtained by this method are not the usual ones. For example, if we take $\theta = p_i dq^i$, the quantum operators are given by:

$$q(q^i) = q^i + i\hbar\frac{\partial}{\partial p_i}, \qquad q(p_i) = -i\hbar\frac{\partial}{\partial q^j} \tag{88}$$

In fact these operators are not a irreducible representation of the canonical commutation relations. It is necessary to apply other techniques (such as polarization) to obtain the correct ones.

Remark: The term $\nabla_{X_F} = \mathcal{L}_{X_F} - \frac{i}{\hbar}\theta(X_F)$ can be understood as a connection on a complex line bundle with $\mathrm{curv}(\nabla) = -\frac{i}{\hbar}\omega$. This will be useful later for the generalization to a multisymplectic formalism, searching for a connection whose $\mathrm{curv}(\nabla) = -\frac{i}{\hbar}\Omega$.

3.2 A geometric approach to quantization of fields

We are going to generalize the previous conditions to observables that can take values in the algebra of forms. My proposal is the following:

1. (D1) If $d\eta = 0$, then $q(\eta) = \eta\mathbb{I}$.
2. (D2) $[q(F), q(G)] = i\hbar\, q(\{F, G\})$.

where $[A, B]$ is the Schouten–Nijenhuis bracket, defined by[3]

$$[A, B] = AB - (-1)^{|A||B|}BA \tag{89}$$

and the Poisson bracket:

$$\{A, B\} = (-1)^{n-1-b}\iota_{X_B}\iota_{X_A}\Omega \tag{90}$$

Again, we are searching for an algebra morphism $q : (obs, \{\cdot,\cdot\}) \to (\mathrm{end}(\mathbb{H}), [\cdot,\cdot])$

Ansatz:

$$q(F) = F - i\hbar\mathcal{L}_{X_F} - \Theta(X_F) \tag{91}$$

You can try with this kind of operators but find that no one fulfils these conditions.

However, we can find a subalgebra of the algebra of observables that can be quantized in the geometric style, but using multivectors instead of vectors:

$$\begin{aligned} q(\phi) &= \phi \\ q(\phi d^{n-1}x_\mu) &= \phi d^{n-1}x_\mu \\ q(p_i^\mu d^{n-1}x_\mu) &= -i\hbar\mathcal{L}_{\partial_\phi} \\ q(p_i^\mu) &= -i\hbar\mathcal{L}_{\partial_\phi\wedge\partial_1\wedge\cdots\wedge\tilde{\partial_\mu}\wedge\cdots\wedge\partial_n} \\ q(d^{n-1}x_\mu) &= d^{n-1}x_\mu \end{aligned} \tag{92}$$

You can prove that these operators fulfil both (D1) and (D2).

This is now better than Kanathikov's approach because we obtain the same operators as him, but we don't need to work on Clifford algebras, but on exterior algebras, which are more natural for field observables.

3.3 No-go theorem for geometric quantization

Theorem / Conjecture: There exists no algebra morphism $q : (obs, \{\cdot,\cdot\}) \to (\mathrm{End}(\Omega), [\cdot,\cdot])$ that satisfies the following two conditions:

(D1) If $d\eta = 0$, then $q(\eta) = \eta$.

(D2) $[q(F), q(G)] = i\hbar\, q(\{F, G\})$.

[3] This bracket satisfies $[P, Q] = -(-1)^{(p-1)(q-1)}[Q, P]$:

Proof: We proceed by contradiction and assume that there exists a map satisfying both properties. Let us start from the following expression for the Jacobiator for $(n-1)-$forms:

$$\mathrm{Jac}(\alpha,\beta,\gamma) = \{\alpha,\{\beta,\gamma\}\} + \{\beta,\{\gamma,\alpha\}\} + \{\gamma,\{\alpha,\beta\}\} = d(\iota_{X_\alpha}\iota_{X_\beta}\iota_{X_\gamma}\Omega) \neq 0 \tag{93}$$

Applying the quantization map to both sides, we get:

$$q(\{\alpha,\{\beta,\gamma\}\}) + q(\{\beta,\{\gamma,\alpha\}\}) + q(\{\gamma,\{\alpha,\beta\}\}) = q(d(\iota_{X_\alpha}\iota_{X_\beta}\iota_{X_\gamma}\Omega)) \tag{94}$$

On the left-hand side we apply (D2), and on the right-hand side (D1):

$$-\frac{i}{\hbar}\big([q(\alpha),q(\{\beta,\gamma\})] + [q(\beta),q(\{\gamma,\alpha\})] + [q(\gamma),q(\{\alpha,\beta\})]\big) = d(\iota_{X_\alpha}\iota_{X_\beta}\iota_{X_\gamma}\Omega) \tag{95}$$

Applying (D2) again to the left-hand side:

$$-\frac{1}{\hbar^2}\Big([q(\alpha),[q(\beta),q(\gamma)]] + [q(\beta),[q(\gamma),q(\alpha)]] + [q(\gamma),[q(\alpha),q(\beta)]]\Big) = d(\iota_{X_\alpha}\iota_{X_\beta}\iota_{X_\gamma}\Omega) \tag{96}$$

However, the Schouten–Nijenhuis bracket does satisfy the Jacobi identity, so the expression reduces to:

$$0 = d(\iota_{X_\alpha}\iota_{X_\beta}\iota_{X_\gamma}\Omega) \tag{97}$$

But this is not true in general, and therefore we reach a contradiction. This contradiction arises from assuming the existence of a quantization map satisfying (D1) and (D2) for $(n-1)-$forms, and hence we conclude that no such map exists.

This proof is, however, not rigorous enough, because we have only considered that $q(\alpha)$ is a degree-1 operator if α is an $(n-1)-$form. For different degree operators (and possibly a combination of different degrees), a graded Jacobi identity is satisfied, so signs has to be taken into account. A full proof is left as an exercise for people smarter than me. □

3.4 Alternative: higher categories and gerbes

The fact that the existence of this map fails due to the Jacobiator suggests that we should not look for an algebra morphism, but rather for a morphism of higher categories.

We recall that the prequantum operator in particle mechanics can be obtained from $q(F) = F - i\hbar\nabla_{X_F}$, where $\nabla_X \equiv \mathcal{L}_X - \frac{i}{\hbar}\theta(X)$ can be understood as a connection on a $U(1)-$principal bundle with $\mathrm{curv}(\nabla) = -\frac{i}{\hbar}\omega$. A natural generalization for multisymplectic formalism is to search for a generalized notion of a principal bundle in higher categories, so that we can find a generalization of a connection $\tilde{\nabla}$ whose curvature is given, not by a $2-$form, but by a $(n+1)-$form, so that it is possible to find $\mathrm{curv}(\tilde{\nabla}) = -\frac{i}{\hbar}\Omega$. As far as I know, this can be done via gerbes. For $n=2$ (i.e. for a multisymplectic $3-$form), a $U(1)-$gerbe is a categorified line bundle, or a "bundle of line bundles", that replaces the transition functions with "transition line bundles", that satisfy certain compatibility condition on triple and quadruple overlaps of the open cover. The analogous of a connection here is a pair $(B_\alpha, A_{\alpha\beta})$, with B_α defined at every patch U_α of an open cover and $A_{\alpha\beta}$ defined at every intersection $U_\alpha \cap U_\beta$. The $3-$curvature can be computed by $F = dB_\alpha$.

The following table has been obtained from [6] and summarizes the geometrical objects used in quantum mechanics and their equivalent higher geometric structures.

	symplectic geometry	2-plectic geometry
degree of differential form	2	3
examples	cotangent bundles	exterior square of cotangent bundles
	coadjoint orbits of Lie groups	compact simple Lie groups
classical field theory		
physical objects	particles	strings
observables	Lie algebra of functions	Lie 2-algebra of Hamiltonian 1-forms
measurement	$x =$ point in phase space $x \mapsto f(x)$	$\gamma =$ path in multiphase space $\gamma \mapsto \int_\gamma \alpha$
prequantization		
prequantum structure	principal U(1)-bundle with connection **or** Hermitian line bundle with connection	U(1)-gerbe with 2-connection **or** 2-line stack with 2-connection
local data for prequantum structure	Deligne 1-cocycle: transition functions, 1-forms	Deligne 2-cocycle: transition functions, 1-forms, 2-forms
infinitesimal symmetries of prequantum structure	Atiyah algebroid (Lie algebroid)	Courant algebroid (Lie 2-algebroid)
quantization		
example	$\mathbb{R}^2 \setminus \{0\}$, $\omega = d\theta$	$\mathbb{R}^3 \setminus \{0\}$, $\omega = dB$
polarization	concentric circles	concentric spheres
Bohr-Sommerfeld condition	$\int_{S^1} \theta \in 2\pi i\mathbb{Z}$	$\int_{S^2} B \in 2\pi i\mathbb{Z}$
quantum states	wavefunctions of harmonic oscillator	representations of SU(2)

4 Annex: sign conventions

$$\iota_{X_F}\Omega = dF \tag{98}$$

$$[A, B] = A \circ B - (-1)^{|A||B|} B \circ A \tag{99}$$

$$\mathcal{L}_{X_F} = d\iota_{X_F} - (-1)^{n-f}\iota_{X_F} d \tag{100}$$

$$[\mathcal{L}_{X_F}, \mathcal{L}_{X_G}] = \mathcal{L}_{[X_F, X_G]} = -\mathcal{L}_{X_{\{F,G\}}} \tag{101}$$

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